TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Specific Heat of a Liquid by the Method of Mixture

Experiment 12 turned inside out: the heat source is now a solid whose specific heat is already known, and the unknown belongs to the liquid it warms.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 13 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the specific heat s_l of a liquid (here turpentine oil) by the method of mixtures. A solid whose specific heat s is known — a brass cylinder, s = 0.09 cal g^{-1} $°C^{-1}$ from Experiment 12 — is heated in steam to θ_2 and dropped into the calorimeter, which this time holds the liquid at θ_1. At the common final temperature θ, the heat the solid surrenders is shared between the liquid and the calorimeter (water equivalent W):

$$m_s\, s\, (\theta_2 - \theta) = m_l\, s_l\, (\theta - \theta_1) + W\, (\theta - \theta_1)$$

$$s_l = [\, m_s\, s\, (\theta_2 - \theta) - W\, (\theta - \theta_1)\,] \,/\, [\, m_l\, (\theta - \theta_1)\,]$$

The method is the mirror image of Experiment 12, and it shows the logic of the whole mixtures family: calorimetry cannot create knowledge from nothing — one substance's thermal constant must already be known (water's, by definition; the solid's, from the previous experiment) before another's can be measured. The liquid should be one that does not evaporate quickly or react with the solid, and the calorimeter stays covered to hold down evaporation losses.

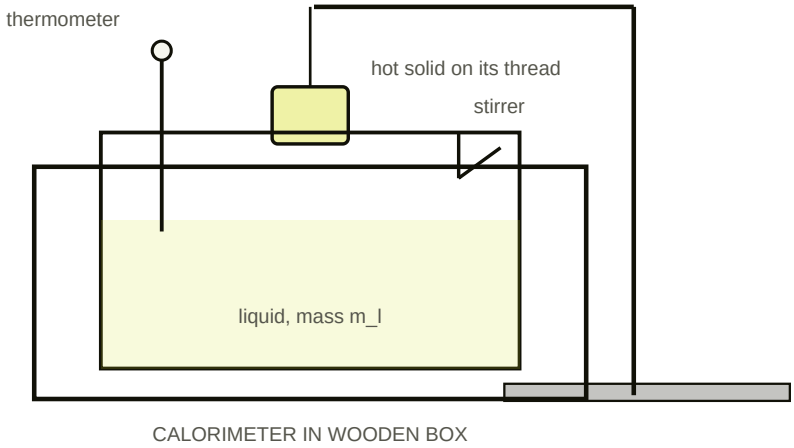


Figure 1 — The mixture, liquid version. The steam-heated solid is lowered into the liquid on its thread — no drop, no splash — and the lid (not shown) goes straight back on to keep evaporation and heat losses down.

2. Apparatus and procedure

- **Apparatus:** calorimeter with stirrer in a wooden box, steam heater, thermometer (0.1 °C), balance, brass solid of known specific heat (s = 0.09 cal g^{-1} $°C^{-1}$) with thread, the liquid under test.
- Weigh the solid m_s and the calorimeter with stirrer (for W). Add a measured mass of the liquid m_l — enough to cover the solid fully — and weigh again if the liquid was added by difference.
- Heat the solid in the steam heater until the heater thermometer is steady at θ_2 = 100 °C.
- Stir the liquid and record θ_1 just before the transfer. Lower the hot solid in gently on its thread, replace the lid, stir, and record the highest temperature θ.
- Separate the calorimeter's share $W(\theta - \theta_1)$ from the heat the solid gave up; the remainder belongs to the liquid. Compute s_l.

3. Worked calculation

Brass solid: m_s = 100 g, s = 0.09 cal g^{-1} $°C^{-1}$, from steam at θ_2 = 100 °C. Turpentine: m_l = 120 g at θ_1 = 28 °C. Calorimeter water equivalent W = 10 g. Final temperature θ = 37.3 °C, a rise of 9.3 °C.

$$\text{heat given out by solid} = 100 \times 0.09 \times (100 - 37.3) = 564.3 \text{ cal}$$

$$\text{calorimeter's share} = 10 \times 9.3 = 93 \text{ cal} \rightarrow \text{liquid receives } 564.3 - 93 = 471.3 \text{ cal}$$

$$s_l = 471.3 / (120 \times 9.3) = 471.3 / 1116 = 0.42 \text{ cal } g^{-1} °C^{-1}$$

In SI units, $s_l \approx 0.42 \times 4180 \approx 1.76 \times 10^3$ J kg^{-1} K^{-1} — the accepted value for turpentine is about 0.42 cal g^{-1} °C^{-1}. Note the bigger temperature rise than in Experiment 12 (9.3 °C against 3.7 °C): gram for gram the oil needs less than half the heat water needs, so the same heat parcel moves its temperature much further.

4. Precautions

- All of Experiment 12's transfer discipline applies — speed, dryness, full immersion, gentle stirring, read the maximum.
- Cover the calorimeter promptly: many test liquids evaporate far faster than water, and evaporation carries heat away invisibly, cooling the mixture and corrupting θ.
- The known value of the solid's specific heat is part of the apparatus: an error in s (say, from a careless Experiment 12) propagates straight into s_l. Calorimetry chains are only as strong as their first link.
- Do not use a liquid that attacks the solid or the calorimeter (acids, for instance) — a chemical reaction is an unbooked source or sink of heat.

Practice (viva preparation)

1. Why must the specific heat of the solid be known before this experiment can begin — and where did that knowledge come from?
2. A 50 g copper ball (s = 0.09 cal g^{-1} °C^{-1}) at 100 °C is dropped into 200 g of a liquid at 30 °C in a calorimeter of water equivalent 10 g. The final temperature is 33 °C. Find the liquid's specific heat.
3. Why does the same hot solid raise the oil's temperature by 9.3 °C but water's by only 3.7 °C in these two experiments?
4. What error follows if the lid is left off and the liquid is a volatile one like ether?
5. Could the hot solid be replaced by an electric heating coil, as in Experiment 7? What would you need to know, and what would replace the "known s"?

Answer key

1. The solid is the measuring instrument here: its heat delivery $m_s\, s\, (\theta_2 - \theta)$ can only be computed if s is already known. It came from Experiment 12, where the same solid was itself the unknown and water — whose specific heat defines the calorie — was the known. One definition and one measurement bootstrap the whole chain.
2. Heat from the ball = 50 × 0.09 × (100 − 33) = 301.5 cal. Calorimeter takes 10 × 3 = 30 cal. Liquid receives 271.5 cal for a 3 °C rise of 200 g: $s_l = 271.5/(200 \times 3) = 0.45$ cal g^{-1} °C^{-1}.

3. Because the liquid's specific heat (0.42) is less than half of water's (1.00): the same heat parcel, spread over a similar mass, buys more than twice the temperature rise. (The masses and the calorimeter's share also differ slightly, but the specific-heat ratio is the main actor.)
4. Evaporation removes the most energetic molecules and their latent heat: the mixture cools by an amount that has nothing to do with the solid's delivery, θ comes out low, and the calculated s_l is too small.
5. Yes — the coil delivers VIt joules, which is Experiment 7's bookkeeping; the "known s of the solid" is replaced by the electrical measurement and the value of J. Either way, the heat input must be certified by something already trusted.

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